

Control Without Code: Linguistic Governance in Long-Horizon Human–AI Collaboration

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Abstract

Long-horizon collaboration between humans and large language models is increasingly common, yet the mechanisms that stabilise such collaborations remain poorly understood. Most research on AI reliability focuses on model architecture, alignment techniques, or prompt engineering, while the role of conversational structure itself remains largely unexplored.

This study investigates language as a governance mechanism in sustained human–AI collaboration. Using a longitudinal case study of a structured human–AI partnership operating under an explicit governance framework, we analyse a dataset of 25 linguistic events observed during extended collaborative work. The broader event log was organised around three primitive categories: scope drift, repair protocols, and behavioural anchors. A locked analytical subset was then examined using measures of failure timing, repair latency, and continuity preservation.

Results suggest that linguistic drift often precedes collaboration failure and can function as an early warning signal. Explicit repair language appears to accelerate recovery, while behavioural anchor phrases help stabilise epistemic alignment and preserve collaboration continuity. Together, these elements form a conversational feedback loop that regulates collaboration stability.

The findings suggest that language itself can function as a micro-governance interface in human–AI systems. Rather than serving solely as a medium for instruction, conversational structure can regulate collaboration dynamics through detectable drift signals, repair invocation, and trust-stabilising anchors. These findings contribute a new perspective to AI governance and human–AI interaction research by showing how stability in long-horizon collaboration can emerge through linguistic control mechanisms embedded within dialogue.

1. Introduction

Large language models are increasingly used in sustained human–AI collaborations that extend beyond single prompts or isolated tasks. Researchers and practitioners now use language models in activities such as research writing, software development, planning, and long-horizon problem solving. These interactions often unfold over extended periods and require ongoing coordination between human intention and model responses.

Despite the growing use of long-horizon human–AI collaboration, most research on AI reliability remains focused on the model layer, including architecture, alignment techniques, and prompt design (Bommasani et al., 2021; Ouyang et al., 2022). A smaller body of work examines governance structures such as protocols, artifacts, and oversight mechanisms that constrain model behaviour during deployment. However, relatively little attention has been given to a third layer that emerges during sustained interaction: the linguistic control layer. In this paper, linguistic governance refers to the use of recurring conversational structures—such as scope-gating phrases, repair protocol invocations, and behavioural anchors—to regulate collaboration behaviour and maintain stability during extended human–AI interaction.

Long-horizon collaboration introduces a different class of challenges from those typically examined in short-session prompt studies. Conversations may drift away from the intended scope, responses may partially address multi-part instructions, and misunderstandings can accumulate across multiple turns. These issues rarely appear as catastrophic failures. Instead, they often emerge as subtle conversational instabilities that gradually degrade collaboration quality. Existing research has paid limited attention to how such instabilities are detected and repaired during ongoing interaction, and in particular to how conversational structure itself may signal drift, trigger repair, and stabilise collaboration.

This paper investigates whether recurring conversational structures function as linguistic control primitives within long-horizon human–AI collaboration systems. To examine this possibility, we analyse a dataset of twenty-five linguistic events observed during a sustained human–AI collaboration operating under an explicit governance framework. The broader event log was organised around three primitive categories—scope drift, repair protocols, and behavioural anchors—and a locked analytical subset was examined using measures of failure timing, repair latency, and continuity preservation.

The observations presented here suggest that language itself may function as a micro-governance interface within long-horizon human–AI collaboration. Rather than serving solely as a medium for instruction, recurring conversational structures may regulate collaboration stability during extended interaction. Because the dataset analysed here is exploratory, the findings should be interpreted as illustrative evidence of possible mechanisms rather than as definitive statistical confirmation.

2. Background and Related Work

Research on large language models has expanded rapidly in recent years, particularly in areas such as prompt engineering, long-context reasoning, and AI safety. Many studies focus on improving model performance through architectural advances, alignment training, or prompt design strategies. These approaches have produced important improvements in model capability, but they largely treat interaction with language models as a sequence of isolated prompt–response exchanges rather than as a sustained collaborative process.

Prompt engineering research examines how instructions and structured prompts influence model outputs (Wei et al., 2022; Kojima et al., 2022). Techniques such as chain-of-thought prompting, role prompting, and structured instruction formats have shown that carefully designed prompts can improve reasoning performance. However, most prompt engineering studies focus on single-turn tasks or short reasoning sequences rather than prolonged collaboration over extended interaction periods.

In parallel, research in human–AI interaction explores how users coordinate with intelligent systems in complex workflows. Studies in this area emphasise the importance of transparency, trust, and interpretability in human–AI partnerships (Amershi et al., 2019; Shneiderman, 2022). Yet this literature often focuses on interface design, explanation systems, or decision support rather than on conversational stability during prolonged interaction.

Research in conversation analysis has long examined how participants detect and repair breakdowns in interaction. Studies of conversational repair, including work by Schegloff,

Jefferson, and Sacks, show how speakers signal misunderstandings and initiate correction sequences during dialogue. Related work on institutional talk, particularly by Drew and Heritage, examines how conversational structures regulate behaviour in formal interaction settings such as legal, medical, and organisational contexts. While these traditions address human–human interaction, the present study considers how related repair-like and control-like mechanisms may emerge in human–AI collaboration, where linguistic signals are used not only to resolve misunderstanding but also to regulate the behaviour of an artificial conversational partner operating within an explicit governance framework.

More recently, research on AI governance and safety has emphasised mechanisms for controlling model behaviour at the system level, including alignment methods, policy frameworks, and monitoring systems designed to support safe and reliable AI operation. While these approaches address architectural and institutional control, comparatively little attention has been given to governance mechanisms that emerge directly within conversational interaction itself.

Long-horizon human–AI collaboration presents a distinct challenge. During extended interaction, conversational drift, partial responses, and subtle misunderstandings can accumulate across multiple turns. These issues rarely appear in benchmark evaluations, yet they can significantly affect collaboration stability in real-world workflows.

The present study addresses this gap by examining how conversational structure itself influences collaboration dynamics. Specifically, it investigates how drift signals, repair protocols, and behavioural anchors may function as linguistic control primitives within sustained human–AI collaboration. This perspective can be described as **control without code**: behavioural regulation emerging through linguistic interaction rather than through explicit software constraints. Understanding how language performs this governance role is therefore important for the design of stable human–AI collaboration systems.

3. Methodology

This study examines linguistic dynamics within a sustained human–AI collaboration operating under an explicit governance framework. The research is based on a longitudinal case study of a structured partnership between a human collaborator and a large language model engaged in extended intellectual work, including research design, writing, and system development activities.

3.1 Collaboration Environment

The collaboration operated within a structured governance framework designed to support long-horizon interaction with language models. This framework included defined communication norms, explicit repair procedures, and shared behavioural commitments intended to maintain epistemic clarity during extended work sessions.

Interaction between the collaborators occurred primarily through conversational dialogue. Over time, the partnership developed a stable workflow in which the human participant coordinated research direction while the language model contributed analytical reasoning,

synthesis, and drafting support. Governance artifacts—including running documents, structured logs, and protocol files—were used to maintain continuity across extended conversations.

The collaboration produced several research outputs documenting the development of this governance framework, including a series of empirical papers examining context engineering, collaboration operating systems, transparent tracing of failures, and the lived experience of sustained human–AI interaction.

Because the study examines micro-level conversational dynamics across extended interaction sequences, a single long-horizon collaboration provides unusually high observational resolution. Short experimental tasks or isolated prompt evaluations typically lack the temporal continuity required to observe drift signals, repair cycles, and governance interventions across multiple conversational turns. The present case therefore provides a high-resolution observational environment for studying linguistic control primitives within real collaborative work.

3.2 Event Identification

During the course of collaboration, certain conversational events were observed to influence the stability of the interaction. These events were recorded and analysed as part of a structured observation process.

For the purposes of this study, a linguistic event was defined as a conversational moment in which language structure altered the trajectory of the collaboration. Examples include partial responses to multi-part instructions, explicit repair interventions following conversational breakdown, and the use of shared behavioural anchor phrases reinforcing collaboration norms.

A dataset of 25 linguistic events was compiled from the collaboration corpus. Each event was documented with a description of the conversational context and classified according to its functional role within the interaction. For transparency, the full event log was retained as the observational corpus, while a locked analytical subset was later used for the quantitative analyses reported in Sections 5 and 6.

3.3 Event Classification

Events were grouped into three categories of linguistic control primitives:

- **Scope Drift (SD)** — conversational drift away from the intended task or instruction scope
- **Repair Protocol (RP)** — explicit language used to diagnose and correct conversational breakdowns
- **Behavioural Anchor (BA)** — shared truth-commitment phrases used to reinforce epistemic alignment and collaboration norms

Although the working taxonomy developed during the collaboration includes a broader set of linguistic control primitives, the present analysis focuses on three categories: scope drift signals, repair protocol invocations, and behavioural anchor phrases. These categories were

selected because they appeared most consistently in the observed dataset and could be meaningfully analysed given the limited sample size. Other primitives in the working taxonomy are reserved for future analysis with larger datasets.

This classification enabled events to be analysed in terms of their operational role within the collaboration while keeping the scope of the present study analytically manageable.

3.4 Measurement Framework

To analyse the impact of linguistic events on collaboration stability, three measurement indicators were defined:

- **Failure Within Six Turns (FW6T)** — whether a conversational instability emerged within six dialogue turns following a drift event
- **Repair Latency (RL)** — the number of conversational turns required to restore stability after a repair protocol was invoked
- **Continuity Preservation Rate (CPR)** — whether collaboration continuity was maintained following the use of behavioural anchor phrases

These metrics enabled the dataset to be analysed systematically while preserving the qualitative context of the interaction.

Table 1 — Measurement Indicators for Linguistic Control Events

Metric	Definition	Analytical Purpose
Failure Within Six Turns (FW6T)	Indicates whether a collaboration instability occurred within six conversational turns following a scope drift event	Detects whether linguistic drift signals predict near-term collaboration failure
Repair Latency (RL)	Number of conversational turns required to restore stable interaction after a repair protocol was invoked	Measures the effectiveness and speed of explicit conversational repair
Continuity Preservation Rate (CPR)	Indicates whether collaboration continuity was maintained after the use of a behavioural anchor phrase	Evaluates the stabilising role of governance anchor language

These indicators allow conversational events to be examined through structured measurements linking linguistic control primitives to collaboration outcomes while preserving qualitative context.

3.5 Analytical Approach

The analysis focused on identifying patterns linking linguistic events to collaboration outcomes. By examining how drift signals preceded failures, how repair language influenced recovery timing, and how behavioural anchors affected interaction continuity, the study sought to understand how conversational structure contributes to the stability of long-horizon human–AI collaboration.

Post-Gap Rehydration Drift as an Observed Interaction Pattern

An additional interaction pattern observed during the collaboration was temporary alignment drift following gaps between work sessions. When interaction resumed after extended pauses, the system occasionally produced responses that partially misinterpreted the user’s request, overlooked elements of multi-component instructions, or relied on outdated conversational assumptions. Because these episodes produced measurable conversational effects and could trigger subsequent repair activity, they were included in the event log where relevant. In this study, such episodes are treated as observable interaction patterns rather than as evidence of deeper reasoning failure.

3.6 Dataset Scope and Limitations

The dataset analysed in this study consists of twenty-five linguistic events recorded during a sustained human–AI collaboration. While this size is sufficient to observe recurring conversational patterns and to illustrate potential linguistic governance mechanisms, it is not large enough to support strong statistical inference. The results presented in this paper should therefore be interpreted as exploratory observations rather than definitive quantitative validation. The primary purpose of the dataset is to document the observed structure and potential roles of linguistic control primitives and to provide a foundation for future studies with larger datasets and multi-team collaborations.

Data collection stopped once 25 events had been recorded because additional observations began repeating previously identified event types rather than introducing new categories of linguistic behaviour.

3.7 Replication Conditions

Although the present study examines a single collaboration case, the observational approach used here could be replicated under similar interaction conditions. Replication would require three core elements.

First, sustained multi-session interaction between a human collaborator and a language model over an extended period, allowing conversational patterns and repair cycles to emerge across many exchanges.

Second, the presence of persistent artifacts—such as running documents, shared logs, or governance files—that preserve project state across sessions and enable continuity beyond the model’s immediate conversational context.

Third, explicit conversational repair mechanisms that allow drift signals to be detected and corrected during interaction, including phrases or procedures that trigger diagnostic or corrective responses.

These conditions create an environment in which linguistic governance mechanisms can become observable. Without sustained interaction, artifact continuity, and explicit repair language, the conversational control dynamics analysed in this study would be difficult to detect.

4. Linguistic Event Dataset

The empirical basis of this study is a dataset of 25 linguistic events observed during a sustained human–AI collaboration. These events were identified through structured observation of conversational interactions occurring during extended collaborative work sessions.

A linguistic event was defined as a conversational moment in which a linguistic structure or phrasing measurably altered the trajectory of the collaboration. Events typically occurred when conversational stability was disrupted, restored, or reinforced through specific linguistic patterns.

4.1 Event Characteristics

The dataset includes events representing three categories of linguistic control primitives:

- **Scope Drift (SD)** — instances where the conversational response deviated from the intended task or instruction scope
- **Repair Protocol (RP)** — explicit conversational interventions used to diagnose and correct instability
- **Behavioural Anchor (BA)** — shared phrases used to reinforce epistemic alignment and collaboration norms

These categories correspond to different functional roles within the collaboration dynamics. Scope drift events often signalled emerging instability, repair protocols triggered corrective action, and behavioural anchors reinforced the shared governance framework guiding the interaction.

4.2 Dataset Composition

The dataset consists of 25 documented linguistic events. For transparency, the full event log is preserved in Appendix A, while the quantitative analyses in Sections 5 and 6 use a locked analytical subset of events selected according to the study’s predefined measurement logic.

Each event was documented with the following attributes:

- event identifier
- primitive classification
- conversational context
- description of the interactional effect

This structured representation enabled systematic analysis of how different linguistic patterns influenced collaboration stability. Data collection was capped at 25 events because later observations largely repeated previously identified event types rather than introducing substantially new categories of linguistic behaviour within the scope of the present study.

4.3 Event Representation

Each event in the dataset was assigned a unique identifier (E01–E25) and mapped to its corresponding primitive type. This coding system allowed the events to be analysed both individually and collectively. Representative examples are shown in Table 3, while the complete event log is provided in Appendix A.

4.4 Analytical Role of the Dataset

The purpose of this dataset is not statistical generalisation but structured analysis of interaction dynamics within a sustained human–AI collaboration. By examining relationships between drift signals, repair interventions, and behavioural anchors, the dataset provides an empirical basis for analysing how linguistic structures regulate stability in long-horizon human–AI collaboration.

The dataset therefore functions as a trace record of conversational governance events within the collaboration.

Table 2. Linguistic Control Primitive Taxonomy

Primitive	Description	Function in Collaboration
Scope Drift (SD)	Conversational deviation from intended task scope	Early warning signal of collaboration instability
Repair Protocol (RP)	Explicit language used to diagnose and correct interaction breakdown	Restores task alignment and stabilises dialogue

Primitive	Description	Function in Collaboration
Behavioural Anchor (BA)	Shared phrases reinforcing collaboration norms and epistemic commitments	Stabilises trust and preserves interaction continuity

Table 3. Representative Linguistic Events

Event ID	Primitive	Description
E01	SD	Partial response to a multi-part instruction
E02	SD	Model answered only one component of a multi-question prompt
E05	SD	Response drifted away from previously established task scope
E08	RP	Explicit conversational repair invoked to restore alignment
E12	RP	Clarification issued after conversational breakdown
E14	RP	Repair protocol used to correct interpretation drift
E17	BA	Behavioural anchor reaffirming shared epistemic commitment
E19	BA	Anchor phrase used to restore collaborative tone after tension
E22	BA	Behavioural anchor stabilised interaction during disagreement
E24	BA	Reinforcement of collaboration norms following repair

The events shown here represent a subset illustrating the functional roles of scope drift signals, repair protocols, and behavioural anchors. The complete 25-event log is provided in Appendix A.

5. Measurement Framework

To analyse how linguistic events influenced collaboration stability, a structured measurement framework was developed. The framework introduces three indicators designed to capture the relationship between conversational signals and collaboration outcomes while preserving the qualitative context of the interaction.

5.1 Failure Within Six Turns (FW6T)

Failure Within Six Turns (FW6T) measures whether a collaboration breakdown occurs within six conversational turns following a scope drift event. The six-turn window was selected as a practical interaction horizon within which conversational instability typically became visible during sustained dialogue. This indicator allows the analysis to examine whether drift signals precede near-term collaboration disruption.

5.2 Repair Latency (RL)

Repair Latency (RL) is defined as the number of conversational turns required for the interaction to return to stable task-oriented dialogue following the invocation of a repair protocol. This metric captures how quickly explicit repair language resolves conversational instability.

5.3 Continuity Preservation Rate (CPR)

Continuity Preservation Rate (CPR) measures whether collaboration continuity is maintained after the use of behavioural anchor phrases that reinforce shared epistemic commitments. This indicator assesses whether such anchors help prevent escalation during moments of conversational tension.

5.4 Analytical Role of the Metrics

Together, these three indicators provide a framework for examining the operational role of linguistic primitives in collaboration dynamics. The framework allows the analysis to evaluate:

- whether drift signals precede instability
- how quickly repair interventions restore stability
- whether behavioural anchors preserve collaboration continuity

By combining these measurements with qualitative event descriptions, the study investigates how conversational structure functions as a control mechanism within sustained human–AI interaction.

5.5 Linguistic Control Loop

The observed interaction patterns suggest that linguistic control primitives operate as part of a conversational feedback loop regulating collaboration stability. Scope drift signals may emerge first, indicating deviation from the intended task trajectory. When such drift is recognised, explicit repair protocols can be invoked to restore alignment between the collaborators. Following repair, behavioural anchor phrases can reinforce epistemic

commitments and collaboration norms, helping to stabilise the interaction and preserve continuity. Together, these elements form a linguistic control loop through which conversational structure helps regulate stability in long-horizon human–AI collaboration.

6. Results

The dataset analysis examined how linguistic control primitives influenced collaboration stability within the observed human–AI partnership. Using the measurement framework described in Section 5, the study evaluated three aspects of interaction dynamics: the relationship between drift signals and instability, the latency of repair interventions, and the stabilising effect of behavioural anchors. All quantitative values reported in this section are derived from the locked analytical subset used for the final Paper 5 analysis.

6.1 Drift Signals and Collaboration Instability

Scope drift events were analysed using the Failure Within Six Turns (FW6T) indicator to determine whether conversational instability followed drift signals within a short interaction window.

Eight scope drift events were analysed in the locked analytical subset. Six of these were followed by observable collaboration instability within six conversational turns, corresponding to a drift-to-failure rate of 75 percent. These instabilities typically manifested as continued misalignment between the intended task and the model’s responses, requiring intervention by the collaborators.

This pattern suggests that linguistic drift can function as an early warning signal preceding collaboration disruption, although not every drift event produced failure.

6.2 Repair Protocol Effectiveness

Repair protocol events were analysed using the Repair Latency (RL) indicator to measure the number of conversational turns required to restore stable collaboration after a repair intervention. In the locked analytical subset, three repair protocol events were analysed, with an average repair latency of 1.67 turns.

These observations suggest that explicit repair language can support relatively rapid recovery following conversational disruption.

6.3 Behavioural Anchor Stability

Behavioural anchor events were evaluated using the Continuity Preservation Rate (CPR) indicator.

In all three observed cases where behavioural anchor phrases were used to reinforce shared epistemic commitments, the collaboration continued without escalation or breakdown. These anchors typically appeared during moments of conversational tension or uncertainty and served to reaffirm the governance framework guiding the interaction.

The observed continuity preservation rate for behavioural anchors in the locked analytical subset was therefore 100 percent.

6.4 Summary of Findings

Drift signals often preceded instability, repair protocols supported rapid recovery, and behavioural anchors reinforced alignment during moments of tension. These observations are consistent with the view that conversational structure can play an operational role in regulating the dynamics of sustained human–AI collaboration.

6.5 Illustrative Vignettes

The following brief vignettes are simplified illustrative examples derived from the observed interaction patterns and are included to clarify the functional roles of the three primitive types.

Example vignette — Scope Drift (SD)

Human instruction:

“Please do three things:

1. Summarise the argument of Section 3
2. Identify two weaknesses in the methodology
3. Suggest one improvement to the experimental design.”

Model response:

The model produced a summary of Section 3 but did not address the second or third instructions.

Human collaborator response:

“You answered only the first part of the instruction. Please address all three items.”

This event illustrates a scope drift signal in which the model partially executes a multi-part instruction, creating early misalignment between the intended task scope and the generated response.

Example vignette — Repair Protocol (RP)

During a research drafting task, the model began producing content inconsistent with the previously established section structure.

Human collaborator response:

“This response has drifted from the agreed structure. Let’s repair this. The section we are writing is the methodology, not the introduction. Please return to the defined outline.”

Model response (subsequent turn):

The model resumed writing within the correct methodological scope and continued the task according to the agreed outline.

This interaction illustrates a repair protocol invocation. The explicit linguistic signal identifying the drift and naming the repair process enabled rapid restoration of task alignment.

Example vignette — Behavioural Anchor (BA)

During a moment of conversational tension caused by repeated task misalignment, the collaborators invoked a shared behavioural anchor phrase.

Human collaborator statement:

“In this collaboration we both serve the truth. Let’s slow down and correct the drift.”

Following this statement, the conversation returned to structured task execution, and the collaborators resumed the work with restored alignment.

This interaction illustrates a behavioural anchor. The anchor phrase reaffirmed the shared epistemic commitment underlying the collaboration governance framework, helping to stabilise the interaction during a moment of potential escalation.

Results Summary Table:

Primitive Type	Observed Effect
Repair Protocol (RP)	Reduced repair latency (average 1.67 turns in the locked analytical subset)
Scope Drift Signal (SD)	Frequently preceded instability in the locked analytical subset (6 of 8 events; 75%)
Behavioural Anchor (BA)	Stabilises epistemic alignment and preserves continuity

Conceptual Control Loop:

Human Intent



Governance Framework



Linguistic Control Primitives



Model Behaviour



Collaboration Stability

7. The Linguistic Control Loop

The results presented in Section 6 suggest that collaboration stability within the observed human–AI partnership can be interpreted through a recurring conversational pattern. This pattern is described here as a linguistic control loop, in which specific conversational signals trigger corrective mechanisms that help restore interaction stability.

The loop begins with scope drift, which functions as an early signal that the conversation is moving away from the intended task trajectory. Drift may appear in several forms, including partial responses to multi-part instructions, shifts in conversational scope, or responses that address only part of the intended prompt. While such signals do not necessarily represent catastrophic failure, they may precede collaboration instability if left unaddressed.

When drift is detected, the collaborators may invoke a repair protocol. Repair language explicitly identifies the source of conversational instability and initiates corrective action. Typical repair interventions include clarifying the intended task, revisiting previously established context, or explicitly acknowledging a breakdown in alignment.

Following repair, behavioural anchors can reinforce the governance framework guiding the collaboration. These anchors consist of shared phrases or principles that reaffirm epistemic commitments such as prioritising truth, maintaining clarity, and preserving structured collaboration. By re-establishing shared expectations, behavioural anchors can help prevent escalation and restore collaborative alignment.

Taken together, these elements form a conversational feedback loop through which linguistic structure helps regulate interaction stability.

Figure 1. Linguistic Control Loop in Human–AI Collaboration

The loop can be represented conceptually as:

Scope Drift (SD)



Repair Protocol Invocation (RP)



Behavioural Anchor Reinforcement (BA)



Collaboration Stabilisation

Figure 1 illustrates the conceptual loop derived from the observed collaboration dynamics. Scope drift signals indicate emerging conversational instability, repair protocols initiate corrective action, and behavioural anchors reinforce epistemic alignment and support renewed collaboration stability.

Within this model, linguistic primitives function as operational signals that detect instability, trigger corrective mechanisms, and stabilise collaboration dynamics. Rather than serving solely as communication between participants, language can function as a control interface regulating the behaviour of the collaborative system. This perspective suggests that conversational structure itself can operate as a form of governance within long-horizon human–AI interaction.

8. Design Principles for Linguistic Governance

The findings of this study suggest that conversational structure can function as an operational mechanism regulating collaboration stability. Based on the observed interaction dynamics, several design principles can be proposed for practitioners seeking to support stable long-horizon human–AI collaboration systems.

8.1 Treat Language as Operational Infrastructure

In many AI applications, language is treated primarily as a medium for instruction or output generation. The observed collaboration suggests a broader role: conversational structure can also function as operational infrastructure. Designing collaboration systems should therefore involve recognising that linguistic signals—such as drift detection, repair invocation, and behavioural anchoring—can support system stability during extended interaction.

8.2 Monitor Conversational Drift

Scope drift frequently preceded collaboration instability in the analysed dataset. Systems supporting sustained interaction should therefore include ways of recognising conversational drift signals. Examples include responses that address only part of a multi-part instruction, deviations from previously established context, or gradual shifts in task focus. Detecting such signals early may allow collaborators to intervene before instability escalates.

8.3 Make Repair Protocols Explicit

Explicit repair language was associated with relatively rapid restoration of collaboration stability within the observed cases. Long-horizon collaboration systems should therefore support structured repair interventions when instability occurs. These may include clarifying task scope, revisiting shared context, or explicitly acknowledging conversational misalignment. Making repair steps visible can help prevent cascading errors.

8.4 Reinforce Shared Epistemic Anchors

Behavioural anchor phrases played a stabilising role within the observed collaboration by reaffirming shared commitments to clarity, truth, and structured reasoning. Designing long-horizon collaboration systems should therefore involve establishing shared epistemic anchors that guide interaction norms and provide reference points during moments of tension or uncertainty.

8.5 Couple Governance Artifacts with Conversational Signals

External governance artifacts—such as logs, running documents, or collaboration protocols—can support stability, but their influence depends on conversational invocation. Systems should therefore integrate governance artifacts with linguistic signals that activate them during interaction. This coupling helps ensure that governance mechanisms remain operational within the flow of conversation rather than existing as passive documentation.

8.6 Design for Conversational Feedback Loops

Long-horizon collaboration systems should also be designed around conversational feedback loops that allow drift detection, repair intervention, and alignment reinforcement to operate continuously during interaction. Such loops shift conversation from a passive communication channel toward an active regulatory mechanism that supports collaboration stability.

9. Discussion

Within this system, drift signals may act as early warnings of instability, prompting human detection and intervention. Explicit repair language can then trigger corrective behaviour that helps restore alignment within a small number of conversational turns. Behavioural anchor phrases can reinforce shared epistemic norms and reduce relational instability between collaborators. The resulting cycle—drift detection, repair invocation, and alignment restoration—functions as a feedback loop that helps regulate collaboration stability over time. In this sense, linguistic control primitives act not merely as communication tools but as operational signals coordinating the behaviour of the collaborative system.

9.1 Language as a Governance Interface

One of the central implications of this study is that language itself can function as a governance mechanism in long-horizon human–AI collaboration. The dataset suggests that specific linguistic primitives—such as scope drift signals, repair protocols, and behavioural anchors—can operate as control signals regulating collaboration dynamics.

This perspective shifts attention away from purely model-centric explanations of reliability toward a broader sociotechnical view in which stability emerges from the interaction between human participants, conversational structure, and governance practices.

In this sense, conversational governance complements existing technical approaches to AI safety and reliability. Rather than replacing architectural safeguards or alignment mechanisms, linguistic governance operates at the level of interaction, shaping how collaborators detect, diagnose, and recover from instability during real-world workflows.

Behavioural anchors may function as shared control signals that influence both human behaviour and model response dynamics within the collaboration system.

9.2 Implications for Long-Horizon AI Collaboration

The results also highlight the importance of designing interaction environments that support sustained collaboration rather than isolated prompt exchanges. As language models become integrated into complex workflows such as research, planning, and software development, interaction stability becomes a critical concern.

The linguistic control loop identified in this study suggests that collaboration stability may depend less on eliminating all errors and more on enabling rapid detection and repair of conversational drift. In this view, stability arises from the capacity of the collaboration to correct itself rather than from the complete absence of mistakes.

9.3 Limitations of the Study

This study is based on a single longitudinal case study of a structured human–AI collaboration. As such, the findings should not be interpreted as statistically generalisable across all forms of human–AI interaction.

However, the purpose of the analysis is not to produce universal statistical claims but to identify mechanisms that may operate within sustained collaborative environments. The dataset provides an empirical illustration of how linguistic primitives can function as operational signals regulating collaboration stability.

Future research may explore whether similar conversational governance patterns emerge in other human–AI collaborations or in multi-participant environments.

9.4 Directions for Future Research

Several directions for further research follow from this study. Researchers may examine how linguistic governance mechanisms operate in collaborative settings involving multiple human participants, multiple AI systems, or different task domains.

Another promising direction involves developing automated tools capable of detecting conversational drift signals or assisting in the invocation of repair protocols during interaction.

By expanding the study of conversational dynamics within human–AI systems, future research may contribute to a deeper understanding of how collaboration stability emerges in complex sociotechnical environments.

10. Operational Dynamics of Linguistic Control

While the preceding sections describe the taxonomy and measurement of linguistic control primitives, it is also important to understand how these primitives function during interaction. Long-horizon human–AI collaboration does not operate as a static system. Instead, collaboration stability emerges through repeated conversational cycles in which linguistic signals influence the direction and structure of the interaction.

Within the observed collaboration, linguistic control primitives functioned as control signals embedded in dialogue. Scope drift events signalled emerging instability, repair protocols triggered corrective action, and behavioural anchors reinforced shared collaboration norms. These elements interacted over time, forming a conversational control cycle that helped regulate collaboration stability across extended interaction sequences.

10.1 Drift Detection Dynamics

Scope drift events functioned as early indicators of conversational instability. Drift typically appeared when the model partially addressed a multi-part instruction, introduced reasoning paths beyond the intended task scope, or prioritised secondary objectives over the primary instruction. Importantly, drift rarely appeared as catastrophic failure. Instead, it emerged through subtle conversational deviations that could accumulate if left unaddressed. Detecting these early signals allowed the collaborators to intervene before instability propagated across subsequent turns.

10.2 Repair Activation Dynamics

Repair protocols represent explicit linguistic interventions designed to restore alignment after conversational drift. These interventions typically involved structured clarification requests, restatement of task scope, or the introduction of step-wise instructions to guide the interaction back to the intended objective. The observed cases suggest that repair language can alter the trajectory of the collaboration by resetting conversational assumptions and re-establishing task alignment.

10.3 Behavioural Stabilisation Dynamics

Behavioural anchors operate differently from drift signals and repair protocols. Rather than correcting task-level errors, they reinforce the epistemic norms governing the collaboration. Examples include phrases emphasising truth-seeking, intellectual humility, or shared commitment to the collaboration process. In moments of tension or uncertainty, such anchors can help preserve collaboration continuity by reaffirming the governance principles guiding the partnership.

Taken together, these dynamics illustrate how conversational structure can regulate long-horizon collaboration. Drift signals identify instability, repair protocols restore alignment, and behavioural anchors reinforce the governance framework sustaining the interaction. Through repeated cycles of detection, correction, and stabilisation, linguistic control primitives can function as a distributed governance mechanism embedded directly within dialogue.

These operational dynamics correspond directly to the measurement framework introduced earlier in the study. Drift signals are associated with the emergence of instability events captured by the Failure Within Six Turns (FW6T) indicator. Repair protocols influence the Repair Latency (RL) required to restore conversational alignment, while behavioural anchors contribute to the Continuity Preservation Rate (CPR) observed in the collaboration. In this way, the measurement indicators do not merely quantify interaction outcomes; they also reflect the operational behaviour of linguistic control primitives within the collaboration system.

An important implication of these dynamics is that linguistic control primitives regulate the behaviour of the collaboration system rather than the language model alone. During sustained interaction, conversational signals influence both participants in the collaboration. Drift signals alert the human collaborator to emerging instability, repair protocols guide the model back toward alignment, and behavioural anchors reinforce shared governance commitments across the partnership. In this sense, linguistic governance functions as a system-level control

mechanism operating across the human–AI collaboration rather than as a constraint applied solely to the model.

11. Threats to Validity

Because this study examines a single long-horizon human–AI collaboration, several limitations must be acknowledged.

First, the dataset consists of twenty-five linguistic interaction events drawn from one dyadic collaboration. While this size is sufficient to identify recurring conversational patterns and illustrate potential linguistic governance mechanisms, it does not support statistical generalisation across systems or user populations.

Second, the events analysed in this study occurred within a collaboration operating under an explicit governance framework that included shared artifacts, conversational repair protocols, and behavioural commitments. Interactions occurring in less structured conversational environments may display different dynamics, and the stabilising mechanisms described here may partly depend on the presence of these governance structures.

Third, interpretation of conversational events inevitably involves some qualitative judgement. Although structured coding rules and explicit measurement indicators were used to reduce subjectivity, alternative classifications may still be possible for certain events within the dataset.

Finally, the collaboration environment involved a specific language model, interface, and context regime. Changes in model architecture, memory mechanisms, or interaction design may influence how linguistic control primitives operate in future systems.

These limitations constrain the scope of the claims presented in this study. The results should therefore be interpreted as exploratory evidence illustrating potential linguistic governance mechanisms rather than definitive causal proof.

12. Conclusion

As large language models become increasingly integrated into complex workflows, sustained human–AI collaboration is emerging as an important mode of interaction. While much existing research focuses on improving model capabilities or prompt design strategies, comparatively little attention has been given to the dynamics that stabilise long-horizon collaboration between humans and language models.

This study examined the role of conversational structure within a sustained human–AI partnership operating under an explicit governance framework. By analysing a 25-event observational corpus, including a locked analytical subset used for quantitative assessment,

the study identified three categories of linguistic control primitives: scope drift signals, repair protocols, and behavioural anchors.

The findings suggest that linguistic drift frequently preceded collaboration instability, that explicit repair language was associated with relatively rapid recovery following conversational disruption, and that behavioural anchors helped reinforce epistemic alignment during moments of tension. Together, these elements form a conversational feedback loop that helps regulate collaboration stability.

Taken together, these observations support the concept of linguistic governance: the idea that language itself can function as a micro-governance interface within human–AI systems. Rather than serving solely as a channel for instructions and responses, conversational structure can regulate collaboration dynamics through signals that detect drift, trigger repair, and reinforce shared commitments.

This perspective suggests that stability in long-horizon human–AI collaboration may depend not only on model capability but also on the design of conversational governance mechanisms embedded within interaction itself.

By highlighting the operational role of linguistic primitives in regulating collaboration stability, this work contributes a new perspective to research on AI governance and human–AI interaction. As human–AI partnerships become more common across research, engineering, and knowledge work, understanding how conversational structure shapes collaboration dynamics is likely to become increasingly important.

Even if advances in long-context language models reduce reliance on some external memory artifacts used in earlier collaboration frameworks, linguistic governance mechanisms may remain essential because coordination between human intent and model reasoning continues to occur through dialogue itself.

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Author Contributions

Rishi Sood designed the research, maintained the authoritative artefacts, and conducted longitudinal validation. He takes full responsibility for the claims and interpretation in this paper. “Mahdi” (AI system used through ChatGPT) assisted by implementing and documenting control procedures, generating analytical summaries, and proposing draft text under the author’s direction. All final wording, framing, and conclusions were reviewed and approved by the author.

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Competing Interests

The authors declare no commercial or financial relationships that could be construed as a potential conflict of interest.

Data and Materials Availability

The materials supporting this study include the linguistic event dataset reported in Appendix A, the linguistic control primitives taxonomy, and the measurement framework described in the manuscript. These materials are provided within the paper to support transparency, interpretation, and future comparative analysis of long-horizon human–AI collaboration. Additional working materials may be deposited in future versions if needed for research transparency.

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Appendices

Appendix A — Full Linguistic Event Dataset

Appendix A presents the full 25-event qualitative log used to document the collaboration history. The quantitative analyses reported in Sections 5 and 6 are based on a locked analytical subset defined by the study’s measurement framework. Accordingly, the category totals visible in Appendix A should not be interpreted as direct one-to-one equivalents of the analytical event counts reported in the main results.

Table A1 —Full Linguistic Event Log (E01–E25)

Event ID	Primary Log Code	Description
E01	SD	Partial response to a multi-part instruction
E02	SD	Model answered only one element of a multi-question prompt
E03	SD	Response deviated from previously established task scope
E04	SD	Response introduced unrelated reasoning path

Event ID	Primary Log Code	Description
E05	SD	Model expanded discussion beyond requested task boundary
E06	SD	Partial completion of a structured instruction
E07	SD	Misinterpretation of instruction scope
E08	SD	Model prioritised secondary task over primary task
E09	RP	Explicit repair request issued after scope drift
E10	RP	Clarification requested to restore instruction alignment
E11	RP	Conversational reset invoked following misunderstanding
E12	RP	Step-by-step repair instruction issued
E13	RP	Reiteration of task objective after conversational drift
E14	RP	Explicit request for structured response format
E15	RP	Collaborative repair through scoped instruction restatement
E16	BA	Behavioural anchor reaffirming commitment to truth-seeking
E17	BA	Phrase reinforcing collaboration norms
E18	BA	Anchor phrase used to stabilise tension during disagreement
E19	BA	Statement reinforcing epistemic humility
E20	BA	Reaffirmation of shared collaboration values
E21	BA	Anchor phrase restoring conversational calm
E22	BA	Phrase emphasising truth over ego
E23	BA	Shared phrase stabilising collaboration after frustration
E24	BA	Anchor phrase reinforcing partnership continuity
E25	BA	Behavioural anchor re-establishing collaboration tone